

**PROJECT-BASED LEARNING (NAAN MUDHALVAN)
ON
PROFESSIONAL READINESS FOR INNOVATION,
EMPLOYABILITY AND ENTREPRENEURSHIP
PROJECT REPORT**

Team ID	NM2023TMID01933
Project Title	IoT Based Weather Adaptive Street Lighting System

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in partial fulfilment for the award of the degree of

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BACHELOR OF ENGINEERING
COMPUTER SCIENCE AND ENGINEERING



ADHIPARASAKTHI ENGINEERING COLLEGE
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BONAFIDE CERTIFICATE

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CHAPTER-1

INTRODUCTION

1.1 Project Overview

This project, titled “The Internet of Things-based Weather Adaptive Street Lighting System” is intended to reduce energy use and improve street lighting effectiveness. The system flexibly regulates street lighting, leading to significant energy savings, using IoT technology and real-time meteorological data. This system turns on the lights for the portions of the streets that are frequently utilised at night while turning off the lights for the portions of the streets that are rarely used during the day. By utilising Internet of Things (IoT) technology and weather data analysis to optimise energy use and raise safety, it provides an intelligent and effective solution for street lighting.

1.2 Purpose

The purpose of the IoT-Based Weather Adaptive Street Lighting System project is to optimize energy efficiency, reduce costs, enhance safety, and provide user control in street lighting through the integration of IoT technology and weather data. It aims to reduce energy use and boost the effectiveness of street lighting. By constantly adjusting the illumination based on current weather conditions and user needs, the initiative intends to eliminate the inefficiencies in conventional street lighting systems.

The project's key objectives are:

- ✓ Energy Optimisation;
- ✓ Cost Reduce;
- ✓ Safety Enhancement.
- ✓ User Flexibility and Control:

CHAPTER-2

IDEATION & PROPOSED SOLUTION

2.1 Problem Statement Definition

The street light is one of the huge expenses in a Country. In Currently a manual system is used where the light will be made to switched ON/ OFF, the light will be made to switched ON in the evening and switched OFF in the morning. Hence there is a lot of wastage of energy in between the ON/OFF when operated by manually. By this system, it'll automatically reduce the energy consumption and also reduce electricity wastage.

2.2 Empathy Map Canvas



FIGURE 2.1: EMPATHY MAP CANVAS

2.3 Ideation & Brainstorming



FIGURE 2.2: IDEATION AND BRAINSTORMING

2.4 Proposed Solution

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	✓ In between the manually controlled ON/OFF system (ON by the darkness and OFF by daylight), street lights waste a lot of electricity. ✓ Empty roads also use more electricity, which is inefficient.
2	Idea / Solution description	The solution of the given problem can be solved in the way of : ✓ Automatically turning on and off the lights based on the current weather. ✓ Whenever it is not required in some paths, It can turn off the light. It will turn on naturally when it required.
3	Novelty / Uniqueness	✓ It will function based on the surroundings, which helps it save more electricity. ✓ By using an object detection sensor, it will use less energy on streets, roads, and other underused areas. ✓ On your own as well as naturally it can be operated.
4	Social Impact / Customer Satisfaction	✓ When there isn't any movement detected, the system can immediately dim or switch off the lighting to save electricity. ✓ It resulted in more effective and economical city services, enhancing people' quality of life.
5	Business Model (Revenue Model)	✓ The devices can assist in lowering running costs by consuming less energy. ✓ By the Consumption of low power, it control the cash flow. Save the Revenue of the Public or private organisation.

6	Scalability of the Solution	✓ The device will operate in a variety of locations and adjust the brightness of the light depending on the movements density. ✓ And make it simple for the municipality Corporation and the populace to access.
-----------------	---	---

CHAPTER-3

REQUIREMENT ANALYSIS

3.1 Functional Requirement

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through Mobile
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Weather sensors:	The system should be able to gather real- time weather data such as Light Intensity from sensors placed at different locations.
FR-4	Adaptive lighting:	The system should automatically adjust the brightness of the street lights according to the weather conditions.
FR-5	Remote monitoring and control:	This will enable operators to check the status of the lights, detect faults, and take corrective action when necessary.
FR-6	Energy efficiency:	The system should optimize energy usage by turning off lights during the day and reducing their brightness during low traffic hours.

3.2 Non-Functional requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should have a user-friendly interface, allowing the operators to easily monitor and control the street lights.
NFR-2	Security	The system should be secure, ensuring that only authorized personnel can access it and make changes.
NFR-3	Reliability	The system should have redundant components, and it should be able to recover quickly from any failures or faults.
NFR-4	Performance	The system should be able to handle large volumes of data, and it should be optimized to minimize network congestion.
NFR-5	Availability	The system should have a backup power supply in case of power outages, and it should be able to recover quickly from any disruptions.
NFR-6	Scalability	The system should be designed to handle a large number of street lights and sensors, and it should be able to scale as the city grows.

CHAPTER 4

PROJECT DESIGN

4.1 Data Flow Diagrams

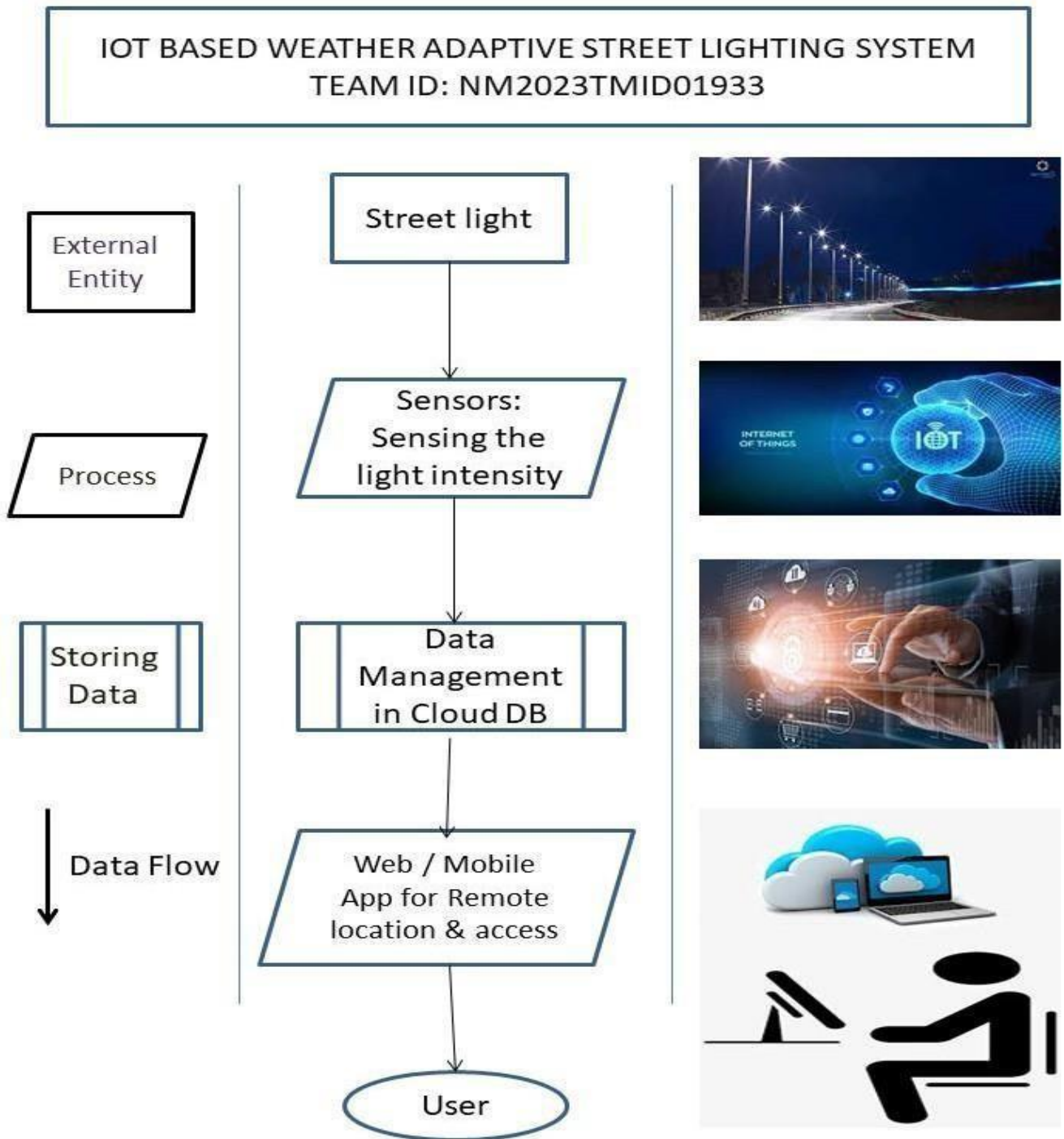


FIGURE 4.1 : DATAFLOW DIAGRAM

4.2 SOLUTION & TECHNICAL ARCHITECTURE

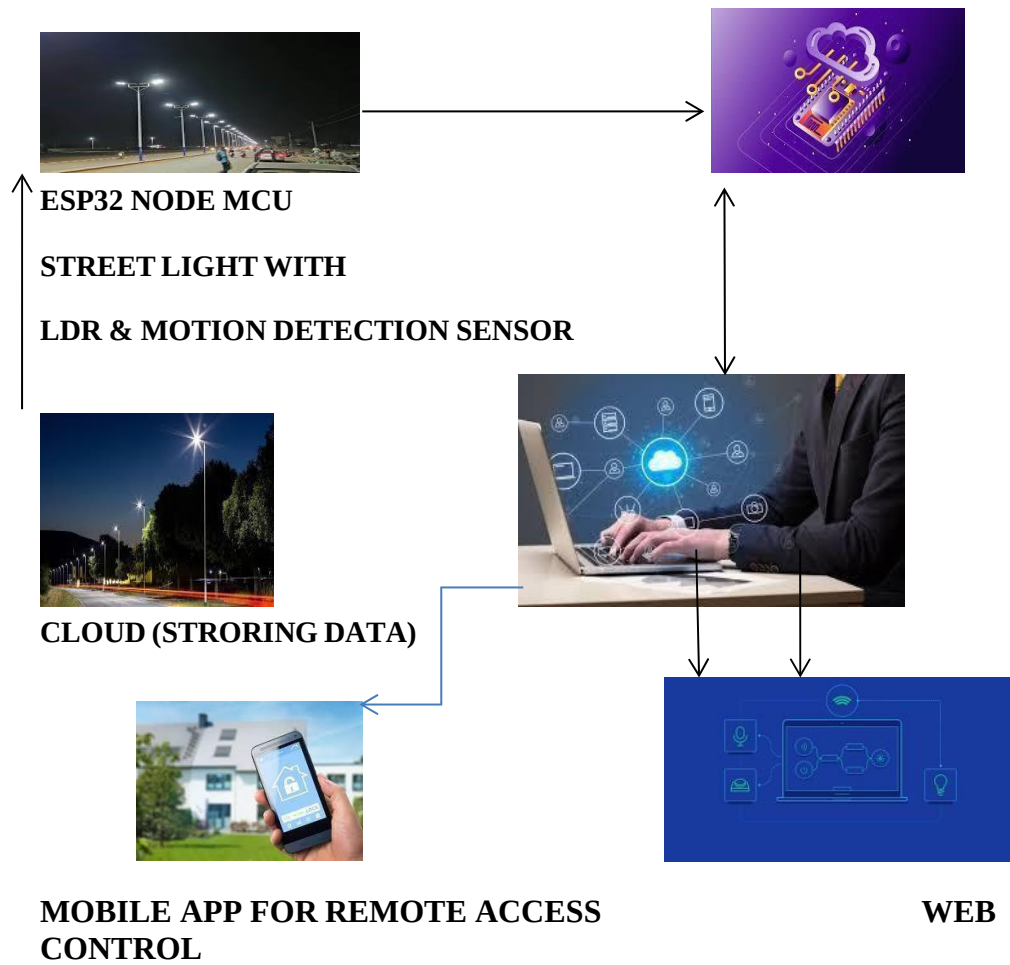


Figure 4.2.1: Solution Architecture Diagram

TECHNICAL ARCHITECTURE

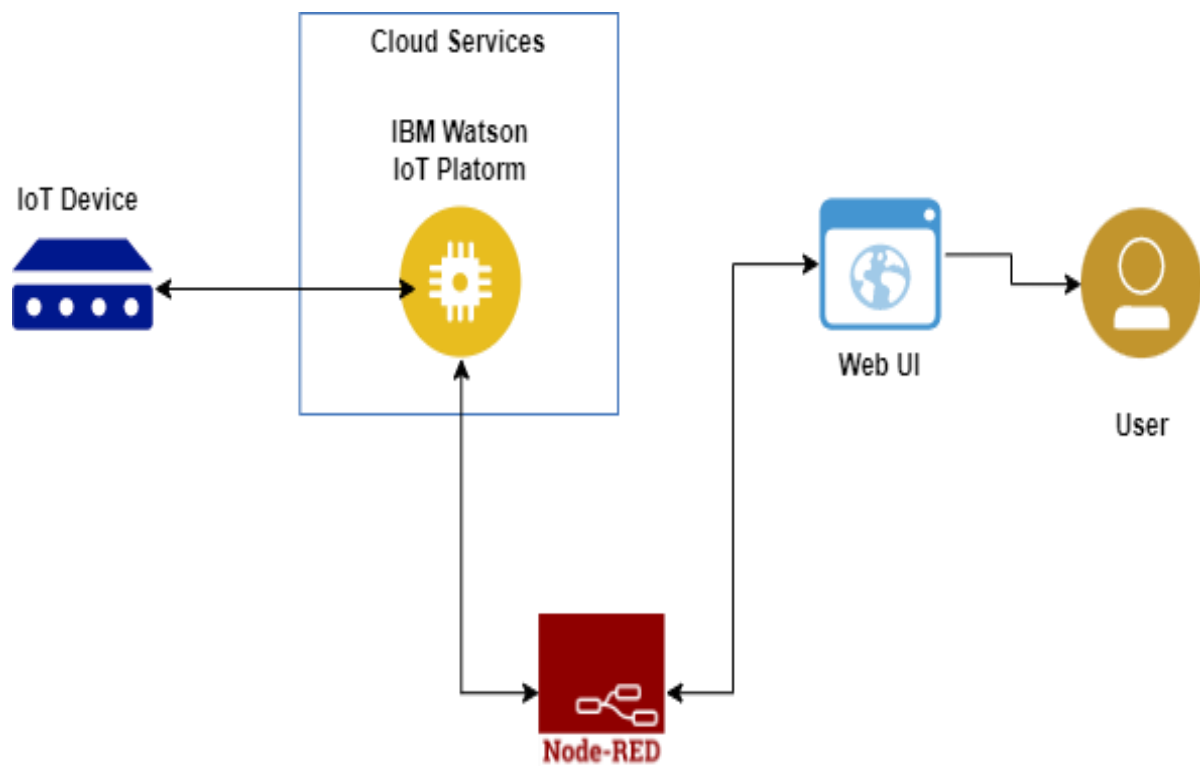


Figure 4.2.2 : Technical Architecture

4.3 User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Team Member
Street light Operator & Maintenance (user)	Web application (Registration)	USN-1	As a user ,I can register for the application by entering my email, password,and confirming my password.	I can access my account /dashboard	High	Harish N
	Login	USN-2	I can log into the application by entering email & password		High	Senthilvel P
System Administrator	Web Application (Dashboard)	USN-3	I want the street lighting system to adapt to changing weather conditions, so that the lighting levels are adjusted accordingly.	The system shall collect real-time weather data, including temperature, precipitation, and ambient light levels.	High	Lokeshkumar V
City Government Official	Web Application	USN-4	I want the street lights to automatically adjustbased on the weather conditions to save energy and improve safety.	The system should provide a dashboard or interface for city officials to monitor the performance and energy savings ofthe weather adaptive street lighting system.	High	Lokeshkumar V
City Planner	Web app (NODE-RED)	USN-5	As a City Planner, I want the street lighting system to adapt to weatherconditions automatically,so that the lighting is optimized for Safetyand energy efficiency.	When the weather conditionchanges to clear skies or high visibility, the street lights should reduce their brightness to conserve energy.	High	Naveen B

CHAPTER-5

CODING & SOLUTIONING

(Explain the features added in the project along with code)

5.1 SOURCE CODE (WOKWI) :

Sketch.ino:

```
#include <WiFi.h> //library for wifi
#include <PubSubClient.h> //library for MQTT

#define IR_PIN 12 // Pin connected to the output of the IR sensor
#define LED 5
#define LED2 2
#define LED3 15
int LDR = 32;
int LDRReading = 0;
int threshold_val = 800;
int LEDBrightness = 0;
int flag=0;

void callback(char* subscribetopic, byte* payload, unsigned int
payloadLength);

//-----credentials of IBM Accounts-----

#define ORG "taies2" //IBM ORGANITION ID
#define DEVICE_TYPE "abcd" //Device type mentioned in ibm watson IOT Platform
#define DEVICE_ID "1234" //Device ID mentioned in ibm watson IOT Platform
#define TOKEN "12345678" //Token
String data3;
float h, t;

//----- Customise the above values -----
char server[] = ORG ".messaging.internetofthings.ibmcloud.com"; // Server Name
char publishTopic[] = "iot-2/evt/Data/fmt/json"; // topic name and type of
event perform and format in which data to be send
char subscribetopic[] = "iot-2/cmd/test/fmt/String"; // cmd REPRESENT command
type AND COMMAND IS TEST OF FORMAT STRING
char authMethod[] = "use-token-auth"; // authentication method
char token[] = TOKEN;
char clientId[] = "d:" ORG ":" DEVICE_TYPE ":" DEVICE_ID; //client id
```

```
//-----
WiFiClient wifiClient; // creating the instance for wificlient
PubSubClient client(server, 1883, callback ,wifiClient); //calling the
predefined client id by passing parameter like server id,portand
wificredential
void setup()// configureing the ESP32
{ try{
  // Try block
  Serial.begin(115200);

  pinMode(LED,OUTPUT);
  pinMode(LED2,OUTPUT);
  pinMode(LED3,OUTPUT);
  delay(10);
  Serial.println();
  wificonnect();
  mqttconnect();

}
catch (const std::exception& e) {
  // Handle the exception
  Serial.println("Exception occurred: ");
  Serial.println(e.what());
}
}

void loop()// Recursive Function
{

  // NATURAL LIGHT INTENSITY READING
  LDRReading = analogRead(LDR);
  Serial.print("LDR READING:");
  Serial.println(LDRReading);

  //READING MOTION DETECTION
  int motionDetected = digitalRead(IR_PIN);
  Serial.print("Motion Detected: ");
  Serial.println(motionDetected);
  Serial.println("CHECKING NATURAL LIGHT IS ENOUGH OR NOT");

  if (LDRReading >threshold_val){
    digitalWrite(LED2, HIGH);
    if (motionDetected) {
      LEDBrightness = map(LDRReading, 0, 1023, 0, 255);
      Serial.print("LED BRIGHTNESS:");
      Serial.println(LEDBrightness);
      analogWrite(LED, LEDBrightness);
    }
  }
}
```



```

        analogWrite(LED2, LEDBrightness);
        analogWrite(LED3, LEDBrightness);
        Serial.println("NOT ENOUGH OF NATURE LIGHT & MOTION DETECTED");
        Serial.println("ALL LIGHTS ARE ON");
        Serial.println();
    }

else{
    analogWrite(LED, 0);
    analogWrite(LED2, 0);
    analogWrite(LED3, 0);
    Serial.println("ENOUGH OF NATURE LIGHT");
    Serial.println("ALL LIGHTS ARE OFF");
    Serial.println();
}

delay(3000);

if (!client.loop()) {
    mqttconnect();
}
}
}

/*****          MQTT PROTOCOL          *****/

void mqttconnect() {
    if (!client.connected()) {
        Serial.print("Reconnecting client to ");
        Serial.println(server);
        while (!!!client.connect(clientId, authMethod, token)) {
            Serial.print(".");
            delay(500);
        }

        initManagedDevice();
        Serial.println();
    }
}

void wificonnect() //function defination for wificonnect
{
    Serial.println();
    Serial.print("Connecting to ");

    WiFi.begin("Wokwi-GUEST", "", 6); //passing the wifi credentials to establish
the connection
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
    }
}

```

```

        Serial.print(".");
    }
    Serial.println("");
    Serial.println("WiFi connected");
    Serial.println("IP address: ");
    Serial.println(WiFi.localIP());
}

void initManagedDevice() {
    if (client.subscribe(subscribetopic)) {
        Serial.println((subscribetopic));
        Serial.println("subscribe to cmd OK");
    } else {
        Serial.println("subscribe to cmd FAILED");
    }
}

void callback(char* subscribetopic, byte* payload, unsigned int payloadLength)
{

    Serial.print("callback invoked for topic: ");
    Serial.println(subscribetopic);

    for (int i = 0; i < payloadLength; i++) {
        //Serial.print((char)payload[i]);
        data3 = (char)payload[i];
    }

    Serial.println("data: "+ data3); // BY MANNUAL LEDS ON & OFF THROUGH NODE-
RED API
    if(data3=="lighton1")
    {
        Serial.println(data3);
        digitalWrite(LED,HIGH);
    }

    else if(data3=="lightoff1")
    {
        Serial.println(data3);
        digitalWrite(LED,LOW);
    }

    else if(data3=="lighton2")
    {
        Serial.println(data3);
        digitalWrite(LED2,HIGH);
    }
}

```

```

}

else if(data3=="lightoff2")
{
  Serial.println(data3);
  digitalWrite(LED2,LOW);

}
else if(data3=="lighton3")
{
  Serial.println(data3);
  digitalWrite(LED3,HIGH);

}

else if(data3=="lightoff3")
{
  Serial.println(data3);
  digitalWrite(LED3,LOW);

}

data3="";

}

```

Diagram.json:

```

{
  "version": 1,
  "author": "Anonymous maker",
  "editor": "wokwi",
  "parts": [
    { "type": "wokwi-esp32-devkit-v1", "id": "esp", "top": 26.98, "left": -
75.23, "attrs": {} },
    {
      "type": "wokwi-led",
      "id": "led1",
      "top": -31.31,
      "left": 61.33,
      "attrs": { "color": "red" }
    },
  ],

```

```

{
  "type": "wokwi-led",
  "id": "led2",
  "top": -33.55,
  "left": 105.39,
  "attrs": { "color": "yellow" }
},
{
  "type": "wokwi-led",
  "id": "led3",
  "top": -32.44,
  "left": 159.91,
  "attrs": { "color": "limegreen" }
},
{
  "type": "wokwi-photoresistor-sensor",
  "id": "ldr1",
  "top": -57.28,
  "left": -126.89,
  "attrs": {}
},
{
  "type": "wokwi-resistor",
  "id": "r1",
  "top": 111.97,
  "left": 40.92,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-resistor",
  "id": "r2",
  "top": 158.38,
  "left": 151.48,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-resistor",
  "id": "r3",
  "top": 142.67,
  "left": 95.25,
  "attrs": { "value": "1000" }
},
{
  "type": "wokwi-pir-motion-sensor",
  "id": "pir1",
  "top": -53.6,
  "left": -208.98,
  "attrs": {}
}

```

```

    }
  ],
  "connections": [
    [ "esp:TX0", "$serialMonitor:RX", "", [ ] ],
    [ "esp:RX0", "$serialMonitor:TX", "", [ ] ],
    [ "led1:C", "esp:GND.1", "black", [ "v118.68", "h-2.37" ] ],
    [ "led2:C", "esp:GND.1", "black", [ "v120.92", "h-100.49" ] ],
    [ "led3:C", "esp:GND.1", "black", [ "v118.69", "h-152.77" ] ],
    [ "esp:D32", "ldr1:A0", "green", [ "h-15.43", "v-84.75", "h131.44",
    "v-26.88", "h1.02" ] ],
    [ "esp:D4", "ldr1:D0", "orange", [ "h31.66", "v-168.39", "h-10.93" ] ],
    [ "ldr1:GND", "esp:GND.1", "black", [ "h4.95", "v206.93" ] ],
    [ "esp:3V3", "ldr1:VCC", "blue", [ "v2.96", "h37.63", "v-232.25" ] ],
    [ "led1:A", "r1:2", "green", [ "v48.34", "h15.15" ] ],
    [ "led2:A", "r3:2", "green", [ "v68.3", "h19.52", "v-2.53" ] ],
    [ "led3:A", "r2:2", "green", [ "v91.39", "h24.62" ] ],
    [ "esp:D5", "r1:1", "green", [ "h0" ] ],
    [ "esp:D2", "r3:1", "green", [ "h0" ] ],
    [ "esp:D15", "r2:1", "green", [ "h0" ] ],
    [ "pir1:GND", "esp:GND.2", "black", [ "v145.15", "h12.3" ] ],
    [ "esp:3V3", "pir1:VCC", "green", [ "v17.38", "h36.08", "v29.33", "h-
228.94" ] ],
    [ "esp:D12", "pir1:OUT", "green", [ "h-92.27", "v-5.31" ] ]
  ],
  "dependencies": {}
}

```

Libraries.txt:

Wokwi Library List

See <https://docs.wokwi.com/guides/libraries>

- ✓ MQTT
- ✓ PubSubClient
- ✓ NewPing
- ✓ PIR
- ✓ ESP8266MQTTClient

5.2 Feature 1 (LIGHT INTENSITY)

```
LDRReading =  
analogRead(LDR);  
Serial.print("LDR  
READING:");  
Serial.println(LDRReading);
```

This type of technology can save energy consumption and maintenance costs by improving the performance of street lights based on current light intensity. It does away with the necessity for manual switching or predetermined schedules since the lights can automatically react to the current lighting conditions. This function reduces energy waste and makes sure that the lights are only on when they are required.

5.3 Feature 2 (MOTION DETECTION)

```
int motionDetected =  
digitalRead(IR_PIN);Serial.print("Motion  
Detected: ");  
Serial.println(motionDetected);  
Serial.println("CHECKING NATURAL LIGHT IS ENOUGH OR NOT");
```

The operational hours of streetlights can be shortened by using motion detection sensors, resulting in longer equipment lifespans. Continuously operating traditional street lights suffer from more wear and tear, requiring regular maintenance and bulb replacement. Reduced maintenance costs and longer-lasting lighting infrastructure are the results of motion detection sensors' contribution to reducing total consumption time.

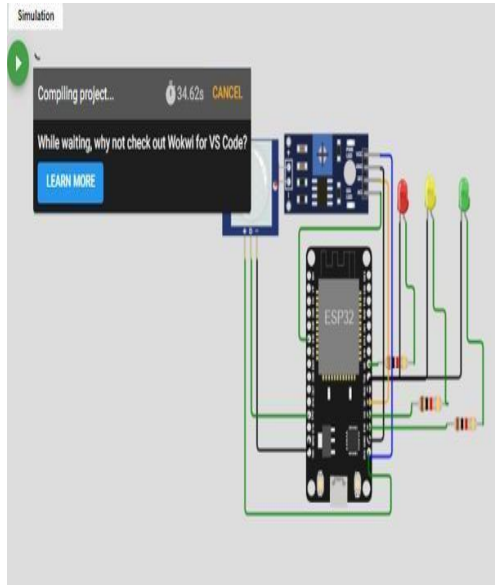
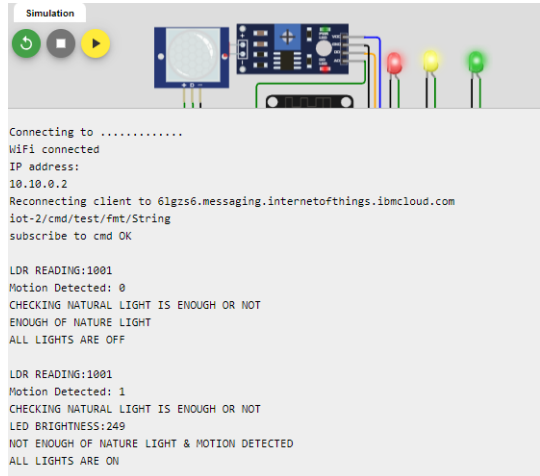
FEATURES ADDED ON THE CODE

```
if (LDRReading >threshold_val){ // LIGHT  
INTENSITYdigitalWrite(LED2, HIGH);  
if (motionDetected) { // MOTION DETECTION  
LEDBrightness = map(LDRReading, 0, 1023, 0,  
255);Serial.print("LED BRIGHTNESS:");  
Serial.println(LEDBrightness);  
analogWrite(LED, LEDBrightness);  
analogWrite(LED2,  
LEDBrightness);analogWrite(LED3,  
LEDBrightness);  
Serial.println("NOT ENOUGH OF NATURE LIGHT & MOTION DETECTED");  
Serial.println("ALL LIGHTS ARE ON");  
Serial.println(); }}
```

CHAPTER-6

RESULTS

6.1 Performance Metrics

Parameter	Values	Screenshot
Metrics	Wokwi Execution time and Output screenshot Or Python accuracy of prediction and output screenshot	EXECUTION TIME : 34.62s  OUT PUT: 

ALL LED S ARE ON :

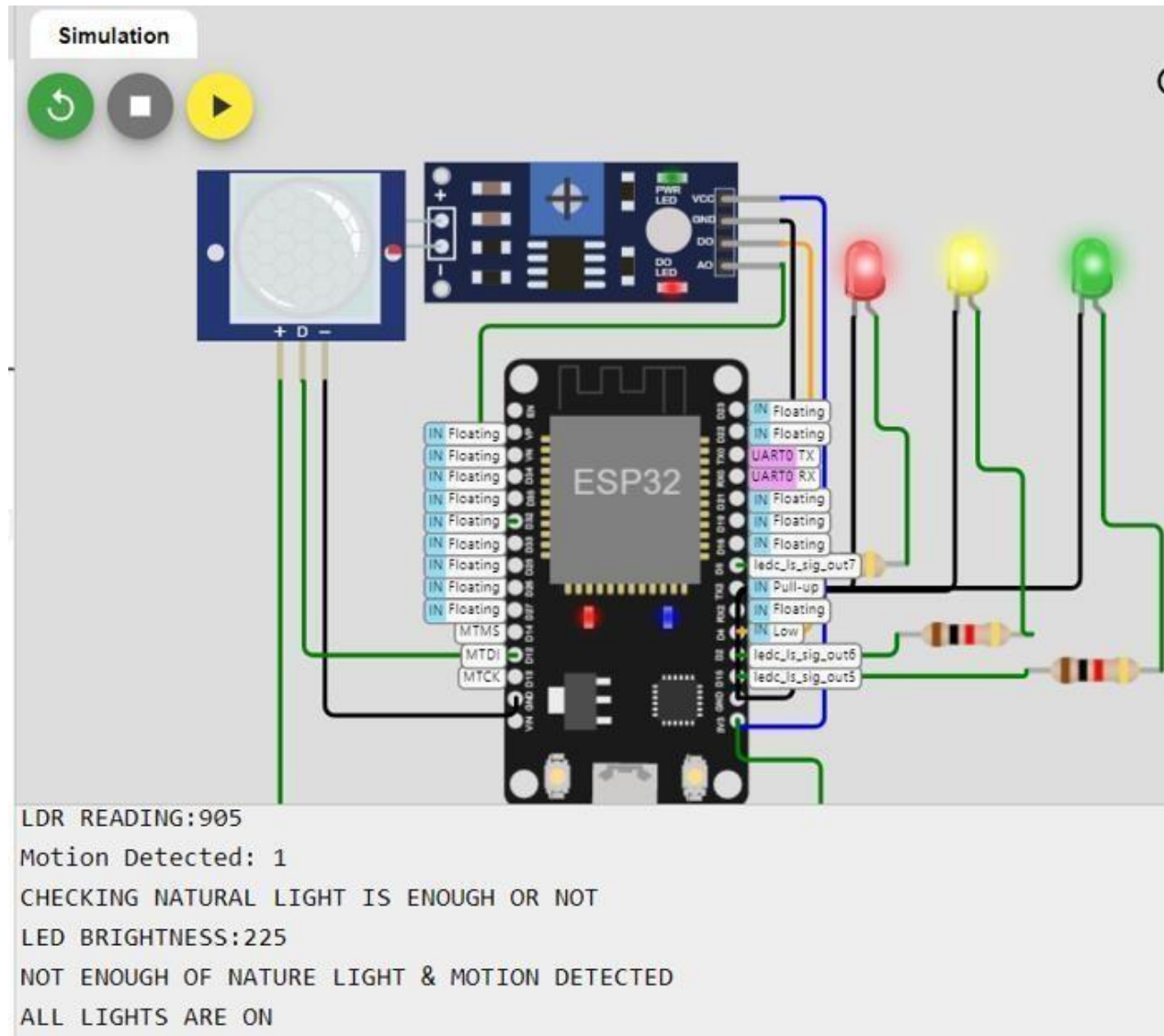


Fig 6.1 Low Natural Light Intensity And Motion Detected

In the above figure 6.1 shows that when Natural Light is not enough for visible and also motion was detected then All LEDs are ON.

ALL LEDs ARE OFF:

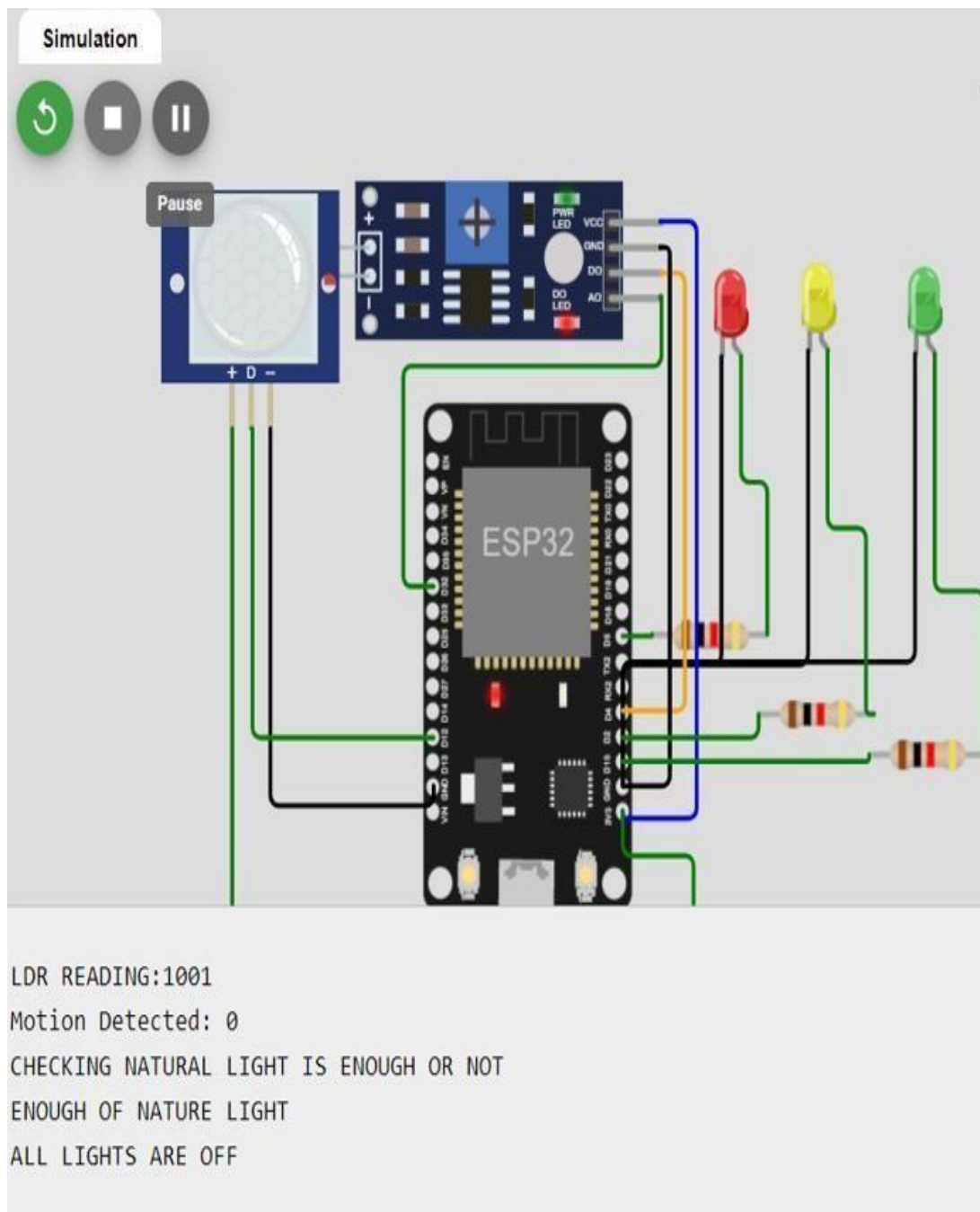


fig 6.2 Natural Light Enough For Visible

In the above Figure 6.2 show that when natural light is enough for visible then All the lights are off.

CONNECTION BETWEEN WOKWI & IBM CLOUD

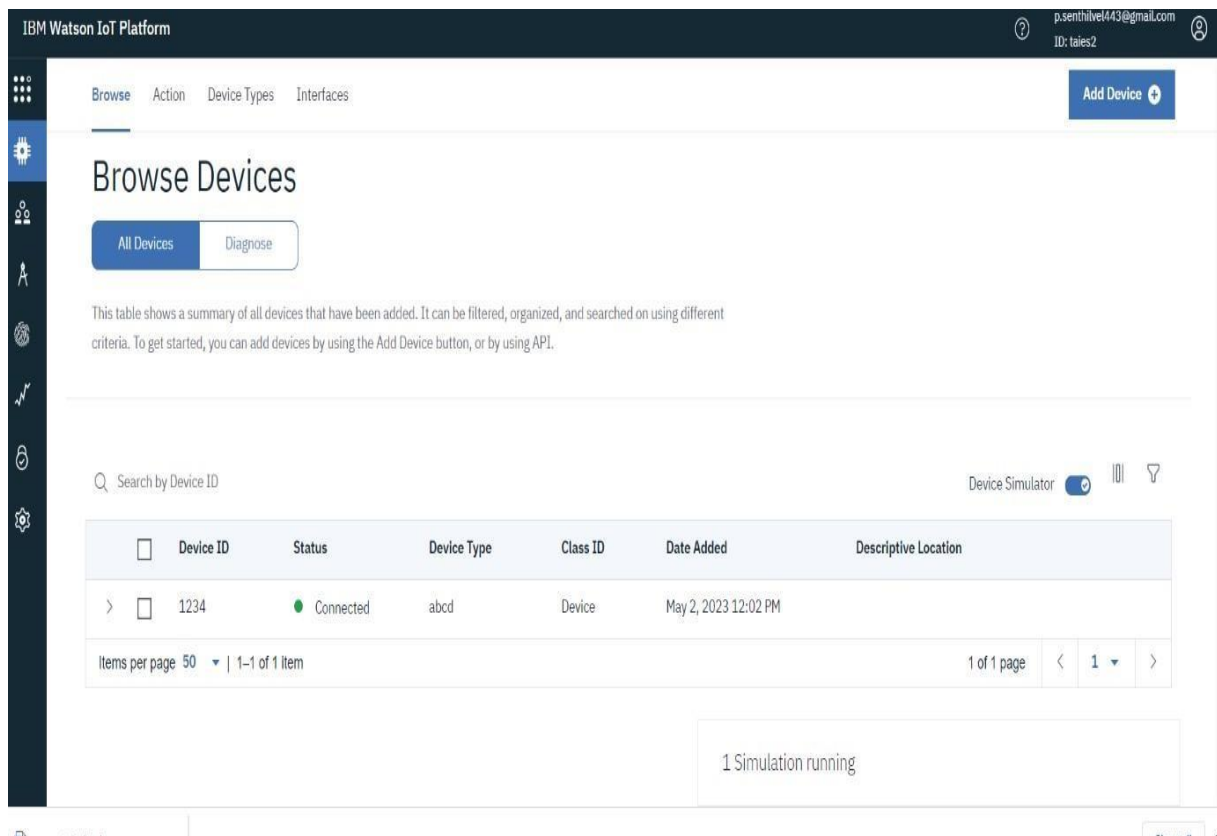


Fig 6.3 IBM WATSON IOT PLATFORM

In the above figure 6.3 it represents the connection between the WOKWI and IBM WATSON IOT PLATFORM, which is used to store data and access anywhere through IBM cloud.

NODE-RED & IBM IOT CONNECTION

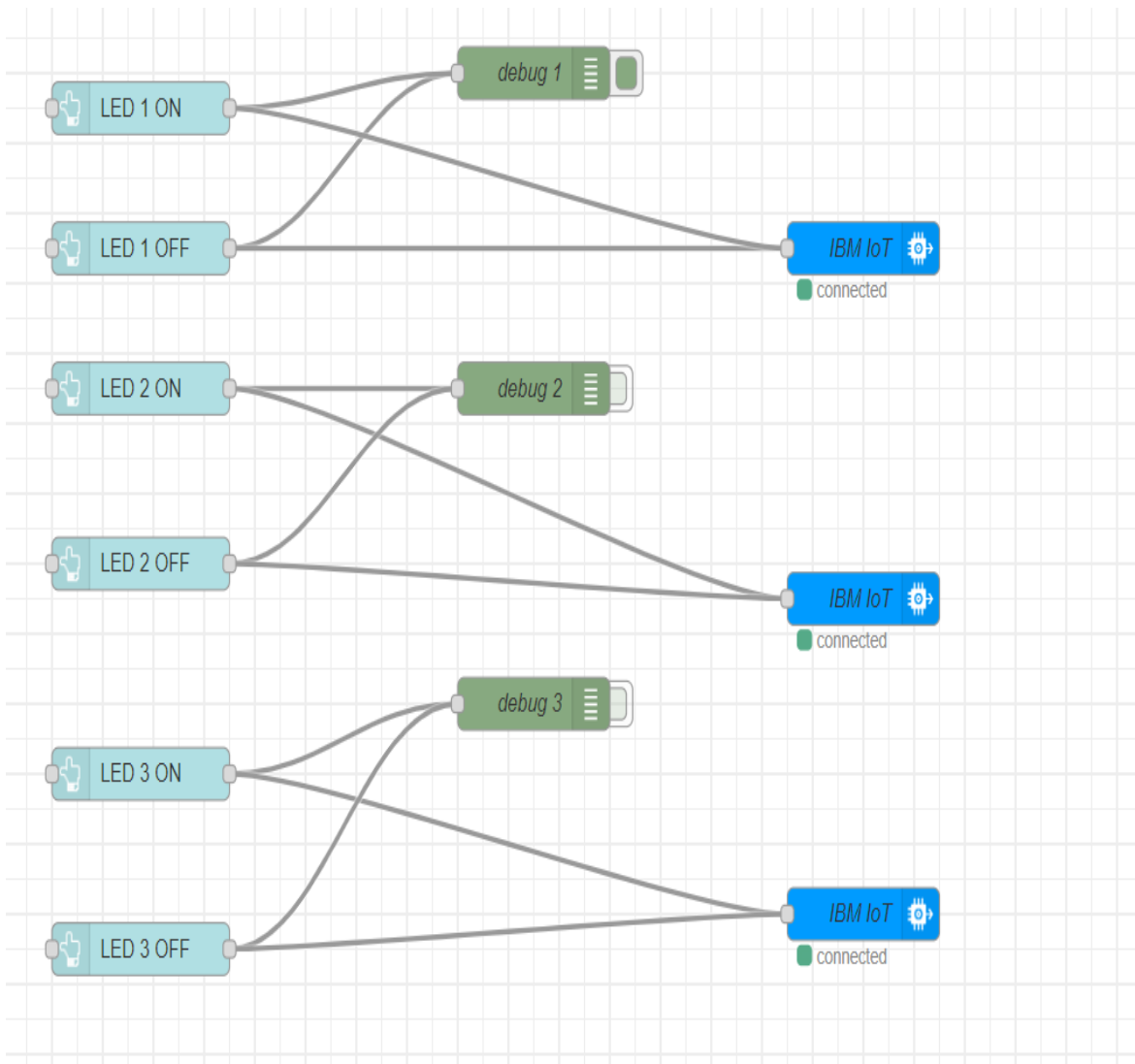


Fig 6.4 CIRCUIT CONNECTION FOR NODE-RED & IBM CLOUD

In the above fig 6.4 it shows the circuit connection of LED LIGHTs control are connected with the IBM IOT platform through the NODE-RED application. .

NODE-RED (USER INTERFACE)

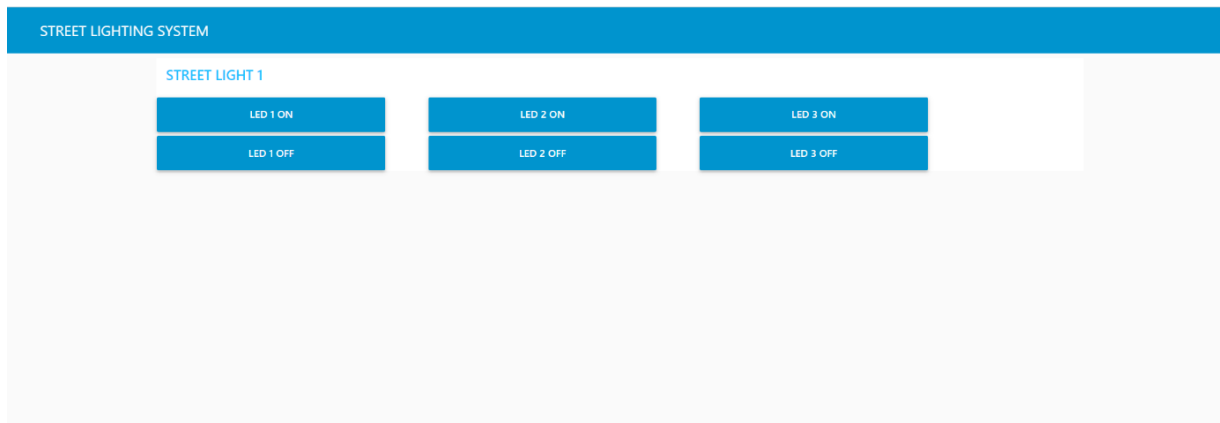


Fig 6.5 NODE-RED application

In the above figure 6.5 shows that the LED lights are controlled manual by NODE-RED user interface (UI) through IBM IOT WATSON.

CHAPTER-7

ADVANTAGES & DISADVANTAGES

ADVANTAGE

- ✓ One Of The Benefits Is Energy Effectiveness.
- ✓ Dependable And Easy To Use Technology:
- ✓ Motion Detection Reduces Electricity Use:
- ✓ Improved Security And Safety:
- ✓ Quick Reaction For Movement

DISADVANTAGE

- ✓ The Initial Expenses For Implementation
- ✓ The Usage Of Connection
- ✓ Data Privacy And Security Issues:
- ✓ Precision And Maintaining Of Sensors:
- ✓ The System Construction Is Difficult

CHAPTER-8

CONCLUSION

To achieve to improve both energy efficiency and safety, the IoT-based weather adaptive street lighting system dynamically modifies lighting settings based on instantaneously weather conditions. It offers useful information for decision-making in urban planning and energy management, enables remote monitoring and control, and encourages preventive maintenance. Overall, it provides better safety, increased energy efficiency, quick maintenance, and insightful data.

CHAPTER-9

FUTURE SCOPE

The future scope of IoT-based Weather Adaptive Street Lighting systems is promising, with potential advancements in energy efficiency, smart grid integration, environmental monitoring, connectivity and communication, data analytics and predictive maintenance, adaptive lighting for safety, integration with smart city infrastructure, and sustainable power sources. These systems can optimize energy consumption, enhance load management, improve urban planning, enable real-time data transmission, support predictive maintenance, enhance safety through adaptive lighting, contribute to smart city initiatives, and incorporate renewable energy sources for sustainability. The overall goal is to create efficient, connected, and sustainable lighting infrastructure that adapts to changing conditions and enhances the quality of life in smart cities.

CHAPTER-10

APPENDIX:

10.1 REFERENCE

1. Street Light Power Management and Control System using IOT Mr. M. Shanmugam¹, K. Mariyappan², M. Meiyazhagan³, B. Jeevanandhan⁴, K. Manikandan⁵ ¹Assistant Professor, Department of ECE, Mahendra Engineering College, Mahendhirapuri, February 2023| IJIRT | Volume 9 Issue 9 | ISSN: 2349-6002 IJIRT 158185 INTERNATIONAL JOURNAL OF INNOVATIVE RESEARCH IN TECHNOLOGY
https://ijirt.org/master/publishedpaper/IJIRT158185_PAPER.pdf
2. . Smart City: Recent Advances in Intelligent Street Lighting Systems Based on IoT Amjad Omar, ¹ Sara AlMaeni,² Hussain Attia,¹ Maen Takruri,¹ Ahmed Altunaiji,³ Mihai Sanduleanu,³ Raed Shubair,⁴ Moh'd Sami Ashhab,^{5,6} Maryam Al Ali,¹ and Ghaya Al Hebsi¹-2022
<https://downloads.hindawi.com/journals/js/2022/5249187.pdf>
3. Research Paper: "Design and Implementation of Weather Adaptive Smart Street Lighting System Using IoT" by S. U. Sayyed, V. S. Bhagwat, and A. B. Ghodake. (2021) Link:
https://www.researchgate.net/publication/349292439_Design_and_Implementation_of_Weather_Adaptive_Smart_Street_Lighting_System_Using_IoT
4. Research Paper: "Smart Street Light Control and Monitoring System Using IoT" by R. S. Lingeshwaran, S. K. Santhosh, and A. Swathi. (2020) Link:
https://www.researchgate.net/publication/342892155_Smart_Street_Light_Control_and_Monitoring_System_Using_IoT
5. Research Paper: "Smart Street Lighting System Using IoT" by S. S. Yadav, K. S. Rakhonde, and A. M. Dhawale. (2020) Link:
<https://ieeexplore.ieee.org/document/9049477>

10.2 GITHUB link

<https://github.com/naanmudhalvan-SI/PBL-NT-GP--5709-1680798923>

10.3 Project Video Demo Link

<https://youtu.be/GCIIdtDwNqUM>



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